radius r as $1/r^2$, and so does gravity's strength. This is *Newton's inverse square law* for gravity.

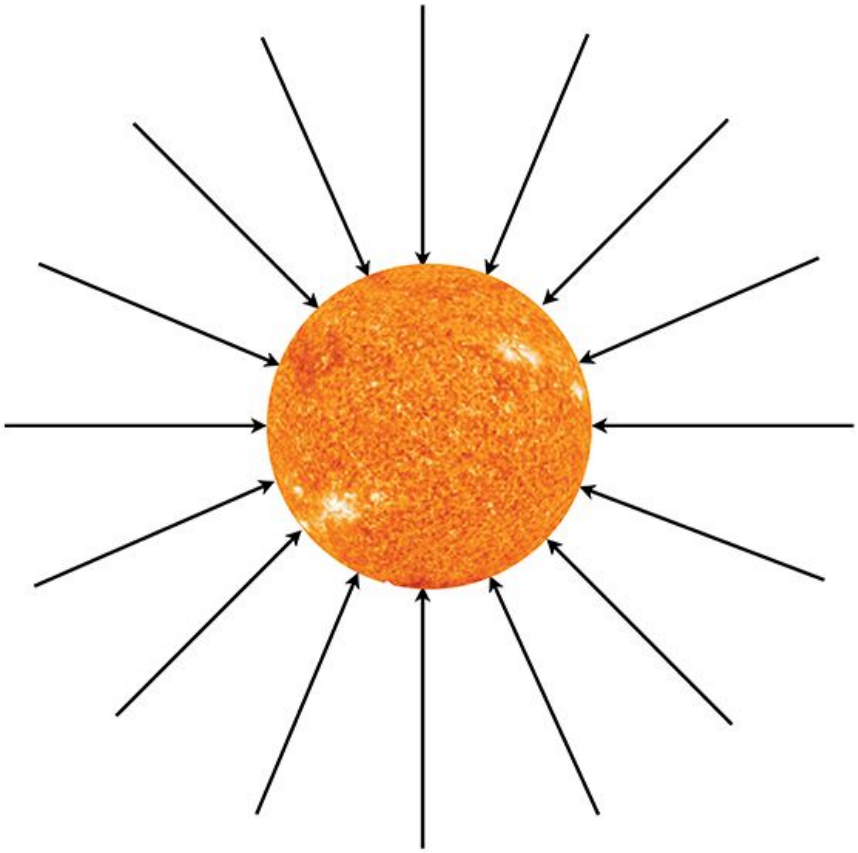


Fig. 23.1. The gravitational force lines around the Sun.

String theory insists that gravity in the bulk is also described by force lines. If space in the bulk is not warped, then the Sun's gravitational force lines will spread radially outward into the bulk (Figure 23.2). Because of the bulk's extra dimension (just one in *Interstellar*), there are three transverse dimensions into which gravity can spread instead of just two. Therefore, *if the bulk exists and is not warped, then the density of force lines and thence gravity's strength should decrease as $1/r^2$ when we move away from the Sun, rather than as $1/r^3$*. The sun's pull on the Earth would be two hundred times weaker, and on Saturn 2000 times weaker. With gravity weakening so rapidly,